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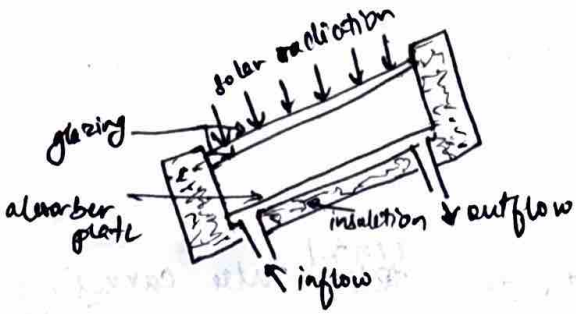
EXCS 'A'

REM
Assignment 2b

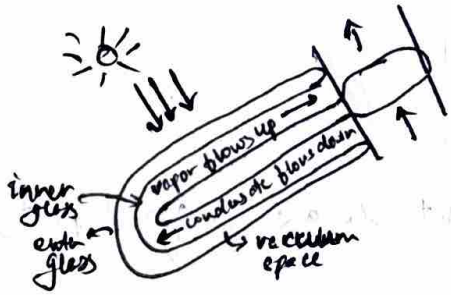
Q1) Draw and explain types of solar collectors

1) Flat Plate Collectors



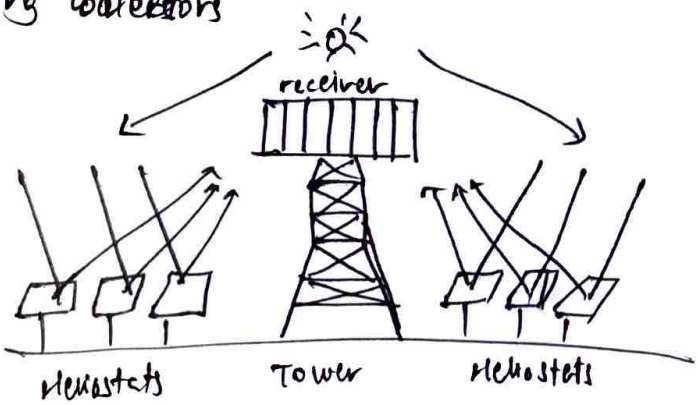
A flat absorber plate captures sunlight and heats fluid (water/air) flowing through tubes under a glass cover.

2) Evacuated Tube Collectors



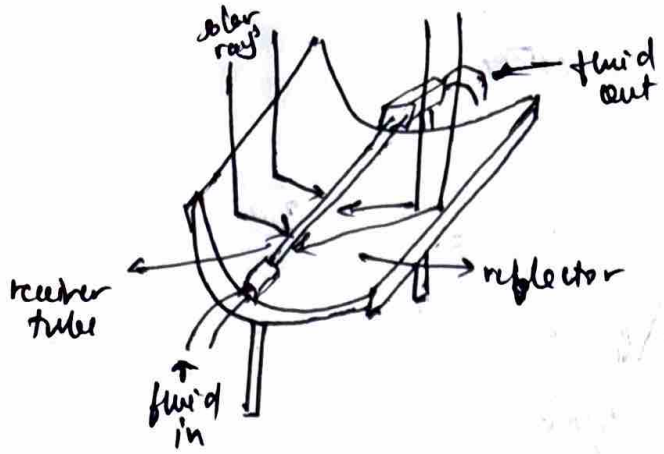
Glass tubes with vacuum insulation reduce heat loss; inner tube absorbs heat and transfers it to fluid

3) Concentrating collectors



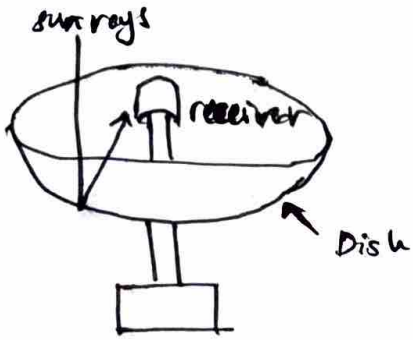
Mirrors / lenses concentrate sunlight onto a small receiver to produce high temperatures.

4) Parabolic Trough collector



curved mirrors focus sunlight onto a central tube carrying heat-transfer fluid.

5) Solar Dish collector



Dish shaped mirror focuses sunlight to a single focal point, often a Stirling engine.

6) Solar tower (Power tower)

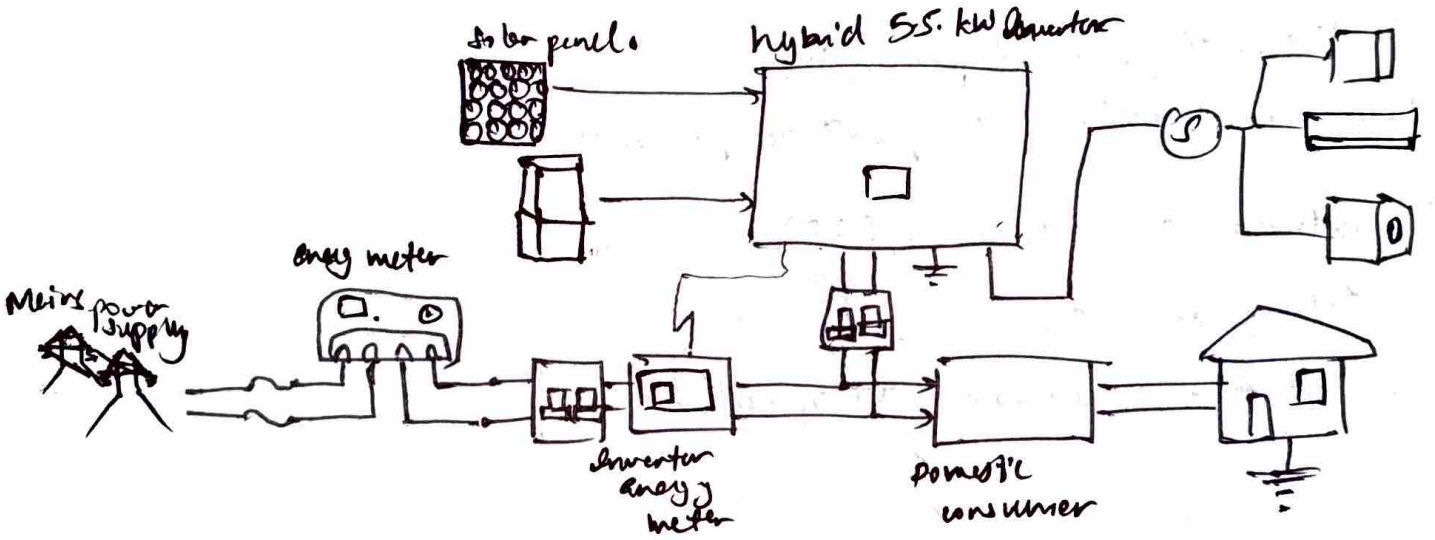
Q2] Explain the working principle of Solar Photovoltaic (PV) systems with a neat diagram. Discuss its applications in India.

Working Principle :

Solar PV system works on the photovoltaic effect, where sunlight falling on a p-n junction semiconductor generates DC electricity.

Main components:

Solar panel
Charge controller
Battery
Inverter
Load / Grid



Applications in India

Rooftop solar
Solar pumps
Street lighting
Rural electrification
Telecom towers

Q3] Explain about 'Solar Thermal Applications' in a global scenario.

Solar water heating

- Domestic and industrial hot water
- Hospitals, hotels, hostels

Solar Cooking

- Box type and parabolic cookers
- reduces LPG consumption

Solar Drying

- Drying agricultural products
- Improves food preservation

Industrial Process Heat

- Textile, dairy, food processing
- Low and medium temperature processes

Concentrated Solar Power (CSP)

- Electricity generation
- Thermal storage enables night operation